



INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP 7.0

Supply Chain Visibility & Optimization

- Turning raw logistics data into real-time, data-driven decisions

Presented by:

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Title: Supply Chain Visibility & Optimization

1.Introduction

"Supply Chain Visibility & Optimization", completed as part of the **Infosys Springboard Virtual Internship 7.0**. This project focuses on analyzing supply chain data and transforming it into meaningful insights that help organizations make faster and more effective business decisions. To understand the project, it is important to first understand the meaning of the title.

A Supply Chain is the complete process involved in delivering a product from the manufacturer to the customer. It begins with purchasing raw materials from suppliers, followed by manufacturing the products, storing them in warehouses, transporting them through various logistics networks, and finally delivering them to customers. Every stage must work efficiently to ensure products are available at the right place, at the right time, and at the right cost.

2.Problem Statement

Modern organizations manage large and complex supply chains involving suppliers, manufacturers, warehouses, transportation, and customers. However, due to limited visibility into these operations, businesses often face challenges such as stock shortages, delayed deliveries, inefficient inventory management, increased operational costs, and poor supplier coordination. These issues reduce overall efficiency and negatively impact customer satisfaction.

The objective of this project is to analyze supply chain data, identify operational inefficiencies, and improve supply chain visibility through data analytics. By preprocessing the data using Python and developing interactive dashboards in Microsoft Power BI, the project provides meaningful insights that support better inventory management, supplier evaluation, logistics planning, and faster, data-driven decision-making.

3.Objectives

The primary objective of this project is to improve supply chain visibility and optimize business operations by analyzing supply chain data and generating meaningful insights. The specific objectives are:

1.To Analyze Supply Chain Data

- Study inventory availability, supplier performance, manufacturing activities, logistics, and customer demand to understand the overall supply chain process.

2. To Preprocess the Dataset

- Clean the raw dataset by handling missing values, removing duplicate records, correcting data types, and preparing high-quality data for analysis.

3. To Develop Interactive Dashboards

- Create user-friendly dashboards in Microsoft Power BI to visualize key performance indicators (KPIs), trends, and supply chain metrics.

4. To Generate Business Insights

- Identify operational inefficiencies and provide data-driven insights that help improve inventory management, supplier evaluation, logistics planning, and overall decision-making.

5. To Improve Supply Chain Performance

- Support organizations in reducing operational costs, minimizing delivery delays, improving resource utilization, and increasing customer satisfaction through better supply chain visibility.

4.Dataset Description

The dataset used in this project contains **1,540 records** and **24 attributes**, representing different stages of the supply chain. It includes information related to products, inventory, suppliers, manufacturing, logistics, transportation, and customer demand. This comprehensive dataset enables the analysis of supply chain operations and supports the development of interactive dashboards for business decision-making.

1. Product Information

This category contains details about the products, including SKU, Product Type, Price, Availability, Products Sold, Revenue Generated, Stock Levels, and Order Quantities. These attributes help analyze product demand, inventory status, sales performance, and revenue generation.

2. Logistics Data

This section includes Shipping Time, Shipping Carrier, Shipping Cost, Transportation Mode, and Route. These attributes are used to evaluate transportation efficiency, delivery performance, shipping expenses, and logistics operations.

3. Manufacturing and Supplier Data

This category contains Supplier Name, Production Volume, Manufacturing Cost, Manufacturing Lead Time, Inspection Result, and Defect Rate. These attributes help assess supplier performance, production efficiency, product quality, and manufacturing costs. Overall, the dataset provides a complete view of supply chain activities, allowing organizations to monitor operations, identify inefficiencies, optimize inventory and logistics, and support data-driven decision-making through analytics and visualization.

5.Data Import & Understanding

1. Import the Dataset

The supply chain dataset was imported into Python using the Pandas library. The dataset was loaded into a DataFrame, which makes it easy to organize, process, and analyze the data. This is the first step before performing any preprocessing or analysis.

2. Examine Structure

After importing the dataset, its structure was examined to understand the number of rows and columns, column names, and data types. Functions such as `df.shape()`, `df.columns()`, and `df.info()` were used to obtain this information and ensure that the dataset was loaded correctly.

3. Review Features

The numerical and categorical features of the dataset were reviewed to understand the information available for analysis. The first few records were displayed using `df.head()`, and `df.describe()` was used to generate statistical summaries of the numerical columns. This helped in understanding the distribution and characteristics of the data.

4. Verify Quality

The quality of the dataset was verified by checking for missing values, duplicate records, incorrect data types, and inconsistent data. Identifying these issues at an early stage ensures that the dataset is clean, reliable, and ready for preprocessing and visualization in Microsoft Power BI.

6.Data Preprocessing

Data preprocessing is the process of cleaning and preparing raw data before analysis. It improves the quality of the dataset and ensures accurate results.

Step 1: Import & Check

- Import the raw supply chain dataset into Python using the Pandas library.
- Load the dataset into a DataFrame.
- Display the first few rows using `df.head()`.
- Check the dataset dimensions using `df.shape()`.
- View column names using `df.columns()`.
- Examine data types and non-null values using `df.info()`.
- Generate summary statistics using `df.describe()`.

Purpose: To understand the dataset and verify that it has been loaded correctly.

Step 2: Identify Issues

- Check for missing (null) values in the dataset.
- Identify duplicate records.
- Verify that each column has the correct data type.
- Detect inconsistent or invalid data values.

Purpose: To identify data quality issues that may affect analysis.

Step 3: Clean & Standardize

- Handle missing values by removing or replacing them.
- Remove duplicate records.
- Correct incorrect data types.
- Standardize data formats for consistency.
- Ensure the dataset is clean and ready for analysis.

Purpose: To improve data quality and prepare the dataset for visualization.

Step 4: Save Output

- Save the cleaned and preprocessed dataset as a new file.
- Import the processed dataset into Microsoft Power BI.
- Use the cleaned data to create interactive dashboards and generate business insights.

Purpose: To provide a reliable dataset for dashboard development and decision-making.

7. Conclusion

The Supply Chain Visibility & Optimization project demonstrates how data analytics can improve supply chain management. By using Python for data preprocessing and Microsoft Power BI for visualization, the project provides clear insights into inventory, suppliers, manufacturing, and logistics. The interactive dashboards help identify operational issues, improve decision-making, reduce costs, and enhance customer satisfaction. Overall, the project shows that data-driven solutions can make supply chain operations more efficient and effective.